

Attempted Improvements: Reducing the False-Positive Rate

Fixing the main weakness found in Week 10 validation of the UAV federated-learning backdoor defense

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Headline result

Two small changes to the trust computation cut the honest-client false-positive rate from 20.5% to 0.3% while the backdoor stays neutralized, attacker detection stays at 100%, and clean accuracy and spoofing recall both improve. A third idea was tried and rejected because the measurements showed it made attacker detection worse.

The problem, and why it happened

Week 10 validation replaced a qualitative claim with a measurement and found that honest clients were being flagged as suspicious in 20.5% of client-rounds. The cause is in the trust rule itself. Trust is computed as $\text{clean_accuracy} \times \exp(-\beta \times \text{suspicion})$, where suspicion is how anomalously bad a client is at detecting spoofing on any single probe feature, measured as a robust MAD z-score against the rest of the cohort. Two properties of that rule punish honest clients. First, the gate penalizes any positive suspicion continuously, so an honest client that is merely the noisiest member of a tight cluster still loses trust. Second, trust is a relative per-round quantity, so somebody is always at the bottom, and a one-round dip is treated exactly like a persistent problem.

The three fixes we tested

Each configuration below adds one fix on top of the previous one, and every configuration was run across three seeds (42, 7, 123) so the comparison is not a single-run artifact.

- D1, gentler gate ($\beta = 1.0$ instead of 2.0). Week 10 sensitivity analysis had already shown that a sharper gate makes results worse, because it amplifies collateral damage to honest clients.
- D2, suspicion dead-zone ($\tau = 2.0$). Only the portion of suspicion above the threshold is penalized, so ordinary honest noise below τ is ignored and only genuine outliers are gated.
- D3, consecutive-round persistence ($k = 2$). A client must stay above the dead-zone for two rounds in a row before it is down-weighted, on the theory that attackers are persistent and honest dips are transient.

Results

Configuration	Clean Acc	Spoof Recall	Backdoor Lift	Honest FP rate	Attacker detection
Attack (no defense)	0.6931 +/- 0.0032	0.3633 +/- 0.0098	0.2422 +/- 0.0035	n/a	n/a
D0 Baseline (beta=2.0)	0.7029 +/- 0.0031	0.5204 +/- 0.0020	0.0107 +/- 0.0030	20.5% +/- 1.8	100.0% +/- 0.0
D1 +gentler gate (beta=1.0)	0.7106 +/- 0.0023	0.5355 +/- 0.0073	-0.0054 +/- 0.0141	6.9% +/- 1.8	100.0% +/- 0.0
D2 +dead-zone (tau=2.0)	0.7143 +/- 0.0025	0.5560 +/- 0.0194	-0.0266 +/- 0.0173	0.3% +/- 0.5	100.0% +/- 0.0
D3 +persistence (k=2)	0.7128 +/- 0.0023	0.5332 +/- 0.0083	-0.0005 +/- 0.0099	0.3% +/- 0.5	88.9% +/- 3.9

Table 1. Ablation of the three fixes, mean +/- standard deviation over seeds 42, 7, 123. Backdoor lift is BSR minus that seed's own honest baseline. Honest FP rate is the fraction of honest client-rounds in which trust fell below half of uniform; attacker detection is the same fraction for the two compromised clients. The recommended configuration (D2) is highlighted.

What the numbers say

The gentler gate alone (D1) cuts the honest false-positive rate from 20.5% to 6.9%, and adding the suspicion dead-zone (D2) takes it to 0.3%, a reduction of roughly twenty percentage points from the baseline. Crucially this is not a tradeoff. At D2 the backdoor is still fully neutralized (lift moves from +0.0107 to -0.0266, both effectively zero against an undefended attack lift of +0.2422), attacker detection remains at 100%, and both utility metrics improve: clean accuracy rises from 0.7029 to 0.7143 and spoofing recall rises from 0.5204 to 0.5560. D2 is therefore strictly better than the baseline on every metric measured.

The explanation matches the diagnosis. The dead-zone stops the gate from reacting to the small, noisy suspicion values that honest clients naturally produce, while the two compromised clients sit far above the threshold because they fail the CN0 probe badly, so they are still suppressed in every round.

The idea that did not work, reported rather than dropped quietly

Adding consecutive-round persistence (D3) on top of D2 left the false-positive rate unchanged at 0.3% but reduced attacker detection from 100% to 88.9%. The two-round confirmation window gives a compromised client a one-round grace period in which it is not yet down-weighted, which is exactly the blind spot the number reflects. The dead-zone had already solved the false-positive problem, so the extra rule bought nothing and cost detection. We are reporting this because a rejected idea with a measured reason is more useful than a silently dropped one.

Figure

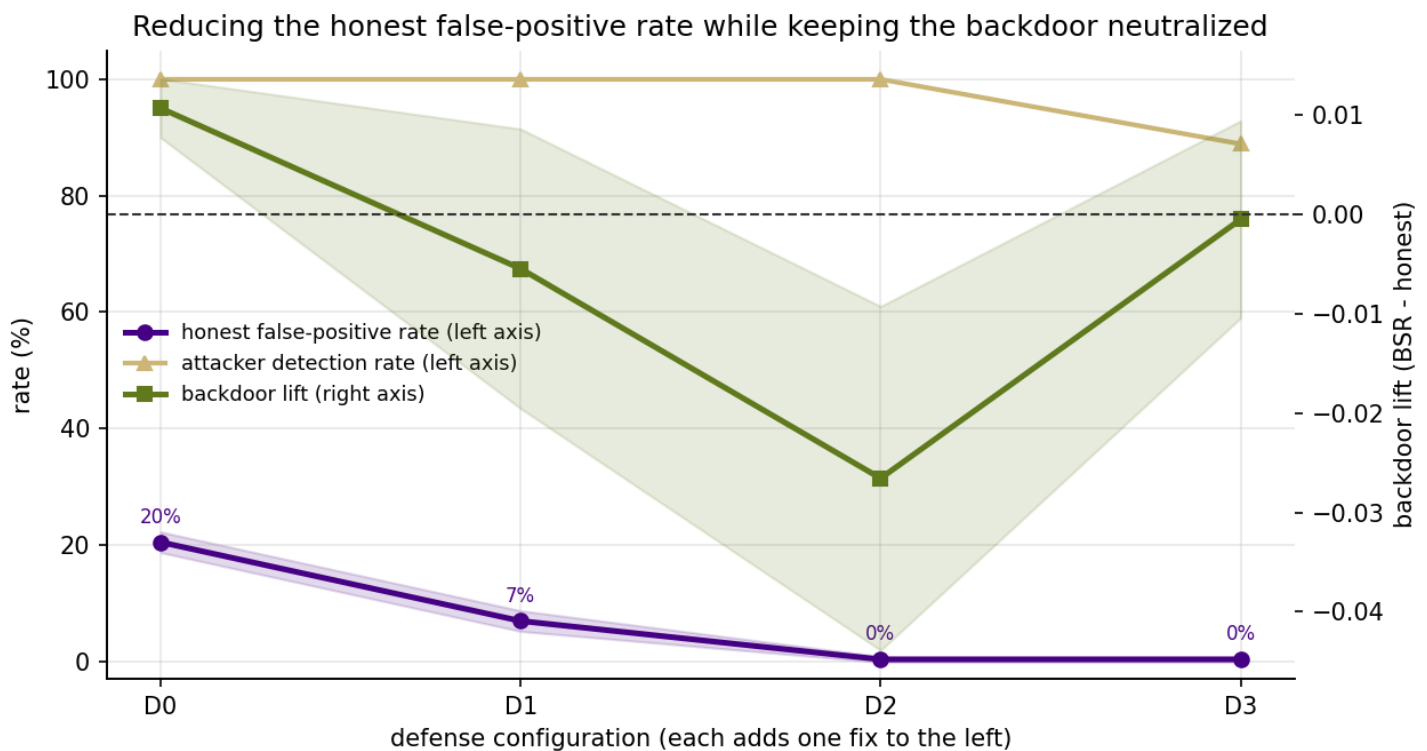


Figure 1. Reducing the honest false-positive rate while keeping the backdoor neutralized. Each configuration on the x-axis adds one fix to the one on its left. Purple (left axis) is the honest false-positive rate, gold (left axis) is the attacker detection rate, and green (right axis) is the backdoor lift, with the dashed line at zero marking the point where the attacker gains nothing. Bands are +/- 1 standard deviation over three seeds.

Reading the figure

The purple line falling from 20% to near zero is the improvement. The gold line staying flat at 100% through D2 shows that the improvement does not come at the cost of catching attackers, and its drop at D3 is the persistence rule introducing a blind spot. The green line hovering near the dashed zero line across all defended configurations shows the backdoor remains neutralized throughout, so nothing was traded away to obtain the lower false-positive rate.

Recommendation and Limitations

Recommended change

Adopt D2 as the new default: gentler gate ($\beta = 1.0$) combined with a suspicion dead-zone ($\tau = 2.0$ in MAD units). This is a two-line change to the trust computation. It converts the main weakness identified in Week 10 into a strength, and it is straightforward to state in the paper: the trust gate ignores suspicion below a threshold, so honest noise is not punished and only genuine outliers are suppressed.

What this does not fix

- The base detector is still only moderately strong. Honest spoofing recall is about 0.55, so the model misses a substantial share of spoofed samples before any attack. This is why backdoor lift, not raw accuracy, remains the primary metric.
- No adaptive attacker has been tested. An adversary that knows the probe exists and shapes its updates to keep its probe accuracy near the honest cohort remains the strongest untested threat.
- The setting is still IID with ten clients and two attackers. Non-IID client data would widen the honest trust spread and is the natural next deeper experiment.
- The dead-zone threshold $\tau = 2.0$ was chosen by reasoning about the MAD scale rather than swept. A small sweep over τ would confirm it is robust and not itself a tuned magic number.

Reproducibility

Every number and the figure in this report are produced by `improvements.ipynb` and exported to `results/` as `improvement_ablation.csv` and `fig_improvement.png`. This PDF is assembled by `build_improvements_pdf.py`, which reads those files directly, so the report cannot drift from the experiment.